

Anisha Bhatnagar

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 github.com/anishabhatnagar  <https://scholar.google.com/citations?user=MoTdTKsAAAAJ&hl=en>

Education

09/2022 – 05/2024 **Masters in Computer Science**, *Courant Institute of Mathematical Sciences, New York University*
New York, USA GPA : 3.806/4

07/2017 – 05/2021 **Bachelors in Technology(Computer Science and Engineering)**, *Amity University Uttar Pradesh, India*
Noida, India CGPA : 8.79/10

Work Experience

08/2024 – Present **Assistant Research Scientist**, *NYU Langone Health*
New York, USA

- Developed **AI-driven solutions for radiology workflows**, achieving **retrieval recall of 98%** by fine-tuning BERT-based retriever/reranker models, supporting precise **Retrieval-Augmented Generation (RAG)** pipelines.
- Integrated **FAISS** to manage a high-dimensional **vector database of 27,000+** radiology reports, ensuring scalable and efficient data retrieval.
- Designed AI systems using models like **OpenAI GPT, Google Gemini, and T5** to generate precise, radiologist-style reports, enhancing accuracy and usability in clinical settings.

09/2023 – 05/2024 **Teaching Assistant**, *New York University*
New York, USA

- Design & Innovation Graduate Course (Fall '23)**, Department of Computer Science
- Natural Language Understanding (Spring '24)**, Center of Data Science.

05/2023 – 08/2023 **Student Researcher (NYU ITP Capstone Project)**, *New York Public Library*
New York, USA

- Engineered a web application for managing 890,000+ digital items at NYPL, enhancing organization and accessibility.
- Implemented **automated asset tagging and metadata generation** for 3 asset types with a **custom Vision Transformer** and **Flan-T5 LLM**, improving search retrieval.
- Enabled **large-scale data processing** and reduced processing time for complex queries by 45% through HPC tunneling.

08/2021 – 06/2022 **Software Development Engineer**, *Accenture*
Pune, India

- Enhanced backend workflows** for insurance applications, focusing on **efficient data processing and reliable deployments**.
- Coordinated **automated testing** during deployments, running **100+ test cases** to ensure **stability and error-free releases**.
- Led weekly production rollouts**, collaborating with cross-functional teams to maintain **high-performance** and ensure uninterrupted service delivery.

04/2020 – 05/2020 **Student Intern**, *AT&T*
Noida, India

- Developed a deep learning **LSTM** pipeline to analyze sentiments in 10,000+ mobile phone reviews, leveraging sentiment features to enhance customer segmentation accuracy to 97.3% using **Spectral Clustering and Random Forests**.

Skills

Programming Languages : Python, C, C++, Java, SQL

Frameworks & Libraries : PyTorch, Hugging Face, Scikit-learn, LangChain, NumPy, Pandas

AI & Machine Learning : Deep Learning, Machine Learning, Artificial Intelligence (AI), Natural Language Processing (NLP), Computer Vision, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Large Vision Models (LVMs), Gemini Pro, OpenAI

Software Development : REST APIs, Data Structures & Algorithms, React, Next.js, Tailwind CSS, Django, Flask, DevOps

Tools & Platforms : Git, Docker, Kubernetes, AWS EC2, Google Cloud Platform (GCP), High-Performance Computing (HPC)

Projects

Hate Content Detection in Videos

- Introduced Hate-LLama, a **multimodal audio-visual language model**, based on LLaMA-7B, finetuned for hate speech detection in online videos, utilizing techniques such as **Data and Model Parallel Training**.
- Hate-LLama analyzes both visual frames and audio to classify hate speech, achieving an **accuracy of 71%**.
- Proposed a benchmark dataset** of 300 videos with 33% hate and 67% non-hate content to address the scarcity of labels.
- Github Link: <https://github.com/anishabhatnagar/Hate-LLaMA> 

Analysis of transformer models on Hindi-English Code-Switched text

- Analyzed performance changes of BERT-style models in sentiment analysis for Romanized code-switched inputs.
- Generated Hindi and English translations, and Hindi transliterations using **GPT-3.5** and IndicXLIT models to support evaluation.
- Evaluated **TwHIN-Bert, mBERT, and XLM-T** in a zero-shot setting, noting a consistent 4-9% performance degradation.
- Github link: <https://github.com/anishabhatnagar/hi-en-senti> 

Autonomous Racing with Reinforcement Learning

- Devised AI agents to play the Trackmania F-1 racing game using the **Soft Actor-Critic (SAC)** algorithm, LIDAR inputs, and RNNs.
- Experimented with pure **LIDAR**, LIDAR with track progress, and hybrid environments to optimize configurations.
- Improved training efficiency with sensory data integration, **achieving a 35-second lap time** near the 30-second human best.
- Github Link: <https://github.com/anishabhatnagar/RL-Racing> 

Publications

03/2021 **A Sentiment Analysis Based Approach for Customer Segmentation**, *Recent Patents on Engineering* 

07/2019 **Machine Learning Techniques to Reduce Error in the Internet of Things**, *IEEE* 
Presented at the 9th International Conference on Cloud Computing, Data Science & Engineering (Confluence), Noida, India, 2019, indexed in IEEE